Imaduddin Sheikh

West Lafayette, IN

+1 (765) 532-1605 | imaduddin.sheikh92@gmail.com | linkedin.com/in/imaduddin-sheikh | github.com/isheikh8492

EDUCATION

Indiana Wesleyan University — **Expected December 2027**
Master of Science in Artificial Intelligence, Data Analytics Specialization — Merrillville IN

Illinois Institute of Technology — **August 2019 — May 2023**
Bachelor of Science in Computer Science — Chicago IL

SKILLS

Programming Languages Python, Java, SQL, C#, C++, R, Bash/Shell, HTML/CSS, JavaScript, VB.NET

Frameworks & Libraries PySide6, PyQtGraph, Qt, NumPy, React, ASP.NET, .NET, WPF

Technologies Git, Azure, Firebase, GCP, Protobuf, Jenkins, pytest, Azure DevOps/TFS

WORK EXPERIENCE

MIFtek Corporation — West Lafayette, IN

Software Engineer III — May 2026 — Present

• Designed the next-generation Arago application in Figma from user stories and realized it in C#/WPF using MVVM, dependency injection, modular views, services, and clear UI/runtime/backend separation.

• Established a reusable WPF design system and control library with shared styling, theming, interaction patterns, and custom controls.

• Engineered the instrument runtime and workflow orchestration system for acquisition, maintenance, and QA/QC operations, managing connectivity, session state, workflow transitions, operator feedback, and native C++ integration through P/Invoke.

• Implemented cross-layer reliability, diagnostics, and test infrastructure, including Serilog logging, exception handling, per-session logs, automated testing, and build integration.

Software Engineer — December 2024 — May 2026

• Developed a modular PySide6/PyQtGraph visualization platform for high-dimensional instrument data, creating reusable views, widgets, workspaces, and UI assets shared across 1D, 2D, multi-channel, live, and archived analysis contexts; sustained 1,000 events per 250 ms across 60 channels and 9 features without missed intervals or queue buildup.

• Added interactive gating and ROI tools with rectangle, ellipse, polygon, and line regions for event isolation, descriptive statistics, color-coded subset visualization, persistence, and downstream analysis.

• Built a state-driven experiment engine coordinating sample sequencing, instrument settings, fluidics, recording, and pause/resume/abort workflows.

• Reduced operator QC review time from ~40 minutes to under 5 minutes by automating validation, statistical classification, historical analysis, visualization, and reporting using silhouette scoring and BGMM clustering.

• Created Protobuf-backed instrument operations tooling for configuration, status, alarms, stale-stream detection, and operator feedback across hardware subsystems.

• Optimized Jenkins CI/CD runtime from ~14 minutes to 2 minutes through dependency caching, parallel pytest execution, and automated reporting.

4835 Studios, Inc. — Chicago, IL

Software Developer Intern — July 2023 — August 2023

• Launched a C#/ASP.NET + React/TypeScript CMS with REST APIs, Tailwind CSS, Azure, and Azure SQL for blogs, contact messages, and job applications.

Americaneagle.com — Des Plaines, IL

Web Application Developer Intern — May 2022 — May 2023

• Shipped .NET e-commerce enhancements using TypeScript, ASP.NET, Entity Framework, VB.NET, LINQ, and SQL Server, and implemented Duo MFA across 40+ client websites to strengthen admin access security.

PROJECTS

DevChronicle
Developer productivity tool in C# and WPF for mining full Git repository history into SQLite, session-based commit analysis, and automated daily summaries; benchmarked local metadata ingestion at 1,153 commits/sec, database writes at 47.6K rows/sec, and checkpointed incremental scans 42% faster across synthetic workloads up to 10K commits and 50K file-stat rows.

Worksheet
Built a C# WPF/ScottPlot visualization pipeline for high-dimensional acquisition records with parameter vectors and waveform captures, sustaining 30,000+ records/sec into a 200K-record rolling buffer with incremental processing and sub-millisecond plot rendering.